3.13 Specific fuel consumption

Specific fuel consumption (SFC) is defined as the amount of fuel consumed by an engine for one unit of energy that is produced. SFC is used to express the fuel efficiency of an IC engine. It measures the amount of fuel required to provide a given power for a given period. It is expressed in Kg/MJ or kg/kW-hr.

Diesel engines have better SFCs than petrol engines. This is because, the Diesel engines have much higher compression ratios (from 16:1 to 20:1) than petrol engines (from 7:1 to 12:1) and hence can convert more of the heat produced into power.

3.14 Simple Calculations in I.C. Engines

The scope of this book will not permit dealing of all the calculations of I.C engines. However some important simple calculations pertaining to simple problems have been dealt.

1. Indicated Power

It is the power produced inside the cylinder and calculated by finding the actual mean effective pressure. The actual mean effective pressure is found as follows.

Let a = Area of the actual indicator diagram, **Sq.cm**

l = Base width of the indicator diagram, **cm**

s = Spring Value of the spring used in the indicator, **N/m²/cm**

If p_m = Actual Mean Effective Pressure, **N/m²**

$$p_m = \frac{sa}{l} \text{ N/m}^2$$

The indicated power of the four–stroke and two stroke engines are found as follows :

(a) *Indicated Power of a Four–Stroke Engine*

Let p_m = Mean Effective Pressure, **N/m²**

L = Length of Stroke, **m**

A = Area of Cross section of the Cylinder, **sq m**

N = RPM of the Crankshaft.

n = Number of cycles per minute.

W = F x D
= P x A x D (∵ P = F/A)

Work produced by the piston / Stroke or / Cycle = [Mean Force acting on the Piston] × [Piston displacement in One Stroke]

$= p_m \times A \times L$ Nm

$= p_m L A$ **Nm**

Work produced by the Piston / min = [Work produced by the Piston/Cycle] × [Number of Cycles / min]

$= p_m L A \times n$

$= p_m L A n$ **Nm/min**

In four stroke I.C. engines, one cycle will be completed in *two revolutions* of the crank shaft. Therefore the work will be produced in every *alternate revolutions of* the crankshaft. Thus the *number of cycles per minute will be equal to half the number of revolutions per minute.*

ie., $n = \frac{N}{2}$

Work produced by the Piston/min $= p_m L A \times \frac{N}{2}$ Nm/min

ie., Indicated Power $= \frac{p_m L A N}{60 \times 2}$ Nm/sec or Joule/sec

$= \frac{p_m L A N}{60 \times 2}$ W

$= \frac{p_m L A N}{60 \times 2 \times 1000}$ **kW**

When p_m is expressed in N/m^2, the indicated power of a Four stroke engine is given by,

$$\textbf{Indicated Power} = \frac{p_m L A N}{60 \times 2 \times 1000} \text{ kW}$$

When p_m is expressed in bar, the indicated power of a Four stroke engine is given by,

$$\textbf{Indicated Power} = \frac{100\, p_m L A N}{60 \times 2} \text{ kW}$$

(b) *Indicated Power of a Two–Stroke I.C. Engine*

In two stroke IC engines, *one cycle* will be completed in *every revolution* of the crank shaft. Therefore the work will be produced in *every revolution of the crank shaft*. Thus, the number of *cycles per minute will be equal to the number of revolutions per minute.*

ie., $n = N$

When p_m is experessed in N/m^2 the indicated power of a Two–Stroke engine is given by,

$$\textbf{Indicated Power} = \frac{p_m L A N}{60 \times 1000} \text{ kW}$$

When p_m is experessed in bar, the indicated power of a Two Stroke engine is given by,

$$\textbf{Indicated Power} = \frac{100\, p_m L A N}{60} \text{ kW}$$

2. Brake Power

The indicated power produced inside the IC engine cylinder will be transmitted through the piston, connecting rod and crank. Since these mechanical parts are moving relative to each other, they will have to encounter resistance due to friction. Therefore a certain fraction of the indicated power produced inside the cylinder will be lost due to friction of the moving parts of the engine. Therefore the net power available at the crankshaft will be equal to the *difference between the indicated power produced inside the engine cylinder and the power lost due to friction*. The net power available at the crankshaft is measured by applying the brake and is therefore called *brake power.*

The amount of the power lost in friction is called *friction power.* The *friction power is the difference between the indicated power and the brake power.*

$$\textit{Friction Power} = \textit{Indicated Power} - \textit{Brake Power}$$

The brake power is calculated as follows :

Let W = Net load acting on the brake drum, **kg**

R = Radius of the brake drum, **m**

N = R.P.M. of the crankshaft

T = Torque applied due to the net load W on the brake drum, **kNm**

$= W \times R$

$$\textbf{Torque} = T = \frac{9.81}{1000} W R \text{ kNm}$$

$\therefore$

$$\textbf{Brake Power} = \frac{2 \cdot NT}{60} \text{ kW}$$

3. Mechanical Efficiency

It is the efficiency of the moving parts of the mechanism transmitting the indicated power to the crankshaft. Therefore it is *defined as the ratio of the brake power and the indicated power.* It is expressed in percentage.

$$\therefore \quad \eta_{Mech} = \frac{Brake\ Power}{Indicated\ Power} \times 100$$

$$= \frac{Indicated\ Power - Friction\ Power}{Indicated\ Power} \times 100$$

$$= \frac{Brake\ Power}{Brake\ Power + Friction\ Power} \times 100$$

4. Thermal Efficiency

It is the efficiency of conversion of the heat energy produced by the actual combustion of the fuel into the power output of the engine. Therefore *it is defined as the ratio of the power developed by the engine to the heat supplied by the fuel in the same interval of time.* It is expressed in percentage

$$\therefore \quad \eta_{Thermal} = \frac{Power\ Output}{Heat\ Energy\ Supplied\ by\ the\ Fuel} \times 100$$

The power output to be used in the above equation may be *brake power* or *indicated power* accordingly the thermal efficiency is called *brake thermal efficiency* or *indicated thermal efficiency.*

*The **brake thermal efficiency** is defined as the ratio of brake power to the heat supplied by the fuel.* It is expressed in percentage.

$$\therefore \quad \eta_{B\ Thermal} = \frac{Brake\ Power}{Heat\ Energy\ Supplied\ by\ the\ Fuel} \times 100$$

i.e.,
$$\boxed{\eta_{B\ Thermal} = \frac{B\ P}{CV \times m} \times 100}$$

Where, m = Mass of the fuel supplied, **kg/s**
CV = Calorific Value of the fuel, **kJ/kg**
BP = Brake Power, **kW**

*The **indicated thermal efficiency** is defined as the ratio of indicated power to the heat supplied by the fuel.* It is expressed in percentage.

$$\eta_{I\ Thermal} = \frac{Indicated\ Power}{Heat\ Energy\ Supplied\ by\ the\ Fuel} \times 100$$

i.e.,
$$\boxed{\eta_{I\ Thermal} = \frac{I\ P}{CV \times m} \times 100}$$

3.15 Simple Problems

Problem 3.1

A single cylinder, two–stroke cycle I.C. Engine has a piston diameter 105 mm and stroke length 120 mm. The mean effective pressure is 6 bar. If the crank shaft speed is 1500 rpm, calculate the indicated power of the engine.

(VTU, March 1999)

Solution :

D = 105 mm L = 120 mm p_m = 6 bar N = 1500rpm N = N (two stroke)

The indicated power of a two stroke engine is given by

$$IP = \frac{100\, p_m\ L\ A\ n}{60} \text{ kW}$$

$$= \frac{100 \times 6 \times 0.12 \times \bullet\ (0.105)^2 \times 1500}{60 \times 4} \text{ kW}$$

$$= \mathbf{15.58\ kW}$$

Problem 3.2

A four–stroke I.C. engine running at 450 rpm has a bore diameter of 100 mm and stroke length 120 mm. The indicator diagram details are ; area of the diagram 4 cm², length of indicator diagram 6.5 cm and the spring value of the spring used is 10 bar / cm. Calculate the indicated power of the engine.

(VTU, Aug/Sep 1999)

Solution :

N = 450 rpm D = 100 mm a = 4 cm² L = 120 mm l = 6.5 cm s = 10 bar / cm

The indicated power of a two stroke engine is given by

- Mean Effective Pressure $p_m = \frac{sa}{l}$

$$= \frac{10 \times 4}{6.5}$$

$$= \mathbf{6.15\ bar}$$

- Indicated power I P $= \frac{100\, p_m\ L\ A\ n}{60}$

$$= \frac{100 \times 6.15 \times 0.12 \times \bullet\ (0.1)^2 \times 450}{4 \times 60 \times 2}$$

$$= \mathbf{2.17\ kW}$$

Problem 3.3

A two–stroke internal combustion engine has a stroke length of 150 mm, and cylinder diameter 100mm. Its mean effective pressure is 5.4×10^5 N/m² and crankshaft speed is 1000, find the indicated power.

Solution :

$L = 150$ mm $\quad D = 100$ mm $\quad p_m = 5.4 \times 10^5$ N/m² $\quad N = 1000$

The indicated power of a two–stroke engine is given by

$$IP = \frac{p_m \; L \; A \; N}{60 \times 1000} \text{ kW}$$

$$= \frac{5.4 \times 10^5 \times 0.15 \times \bullet (0.1)^2 \times 1000}{4 \times 60 \times 1000}$$

$$= \mathbf{10.6 \text{ kW}}$$

Problem 3.4

A six cylinder 4 stroke I.C. Engine develops 50 kW of indicated power at mep of 7 bar. The bore and stroke of the engine cyliner is 70 mm and 100 mm respectively. If the engine speed is 3700 rpm, find the average misfires per minute.

Solution :

Indicated Power developed / cylinder $= \frac{50}{6}$ kW

$$IP = \frac{100 \, p_m \; L \; A \; N}{60 \times 2} \text{ kW}$$

$$\frac{50}{6} = \frac{100 \times 7 \times 0.1 \times \bullet (0.07)^2 \; N}{4 \times 60 \times 2}$$

$$\bullet \quad N = \mathbf{3712 \text{ rpm}}$$

$$\bullet \quad \text{Number of Cycles / min} \quad n = \frac{3712}{2}$$

$$= \mathbf{1856}$$

i.e., Number of Explosions per minute $= 1856$

$$\text{Actual Explosions per minute} = \frac{3700}{2}$$

$$= \mathbf{1850}$$

$$\bullet \quad \text{Number of Misfires} = 1856 - 1850 = 6 \text{ / min}$$

Problem 3.5

Find the indicated power of a four stroke petrol engine. The average piston speed is 70 m per min. The mean effective pressure is 5.5. bar. The diameter of the piston is 150 mm.

Solution :

$$\text{Piston Speed} = 2\,L\,N \text{ m/min}$$

$$= 2\,L \times 2\,n \qquad \left(\because \; n = \frac{N}{2} \text{ for four stroke engine}\right)$$

$$= 4\,Ln$$

$$4\,Ln = 70 \text{ m/min}$$

i.e., $$Ln = 17.5 \text{ m/min}$$

Now $$\text{Indicated Power} = \frac{100\,p_m\,L\,A\,n}{60} \text{ kW}$$

$$= \frac{100 \times 5.5 \times \bullet \times \frac{(0.15)^2}{4} \times 17.5}{60}$$

$$= \mathbf{2.835 \text{ kW}}$$

Problem 3.6

The following readings were taken on a four–stroke I.C. engine :

Diameter of the brake drum = 1.5 m
Diameter of the rope = 10 mm
Load suspended on the brake drum = 100 kg
Spring balance reading = 5 kg
Crankshaft speed = 200 rpm

Determine the brake power of the engine.

Solution :

$$\text{Effective Radius } R = \text{Radius of the Brake Drum} + \text{Radius of the Rope}$$

$$R = \frac{1.5}{2} + \frac{0.01}{2} \text{ m}$$

$$= \mathbf{0.755 \text{ m}}$$

$$\text{Net load on the brake drum } W = (100 - 5) \text{ kg}$$

$$= \mathbf{95 \text{ kg}}$$

$$\text{Torque applied } T = \frac{9.81\,WR}{1000} \text{ kNm}$$

$$= \frac{9.81 \times 95 \times 0.755}{1000}$$

$$= \mathbf{0.7 \text{ kNm}}$$

$$\text{Brake Power} = \frac{2\pi N T}{60} \text{ kW}$$

$$= \frac{2\pi \times 200 \times 0.7}{60} \text{ kW}$$

$$= \mathbf{14.66\ kW}$$

Problem 3.7

A four–stroke single cylinder I.C. engine of 250 mm cylinder diameter and 400 mm stroke runs at a piston speed of 8 m/s. If the engine develops 50 kW indicated power, find its mean effective pressure and the crankshaft speed.

Solution :

$D = 250$ mm $\quad L = 400$ mm $\quad IP = 50$ kW $\quad$ *Piston Speed* $= 8$ m/s

$$\text{Piston Speed} = 2\ L\ N \text{ m/min}$$

$$8 \times 60 = 2 \times 0.4\ N$$

$$N = \mathbf{600\ rpm}$$

The indicated power of a four–stroke engine is given by,

$$IP = \frac{100\ p_m\ L\ A\ N}{60 \times 2}$$

$$50 = \frac{100 \times p_m \times 0.4 \times \pi (0.25)^2 \times 600}{4 \times 60 \times 2}$$

$$p_m = \mathbf{5.09\ bar}$$

Problem 3.8

During the test on a four–stroke diesel engine, the following readings were taken when running at full load.

Area of the indicator diagram	$= 3\ cm^2$
Length of the indicator diagram	$= 5\ cm$
Spring constant	$= 100\ N/cm^2/cm$
Engine crankshaft speed	$= 500$ R.P.M.
Diameter of the engine cylinder	$= 150\ mm$
Stroke of the piston	$= 200\ mm$

Find the indicated power of the engine.

Solution :

$a = 3\ cm^2$ $\quad l = 5$ cm $\quad s = 100\ N/cm^2/cm$ $\quad N = 500$ R.P.M

$D = 150$ mm $\quad L = 200$ mm $\quad = 10^6\ N/m^2/cm$

The mean effective pressure is given by,

$$p_m = \frac{sa}{l} \text{ N/m}^2$$

$$= \frac{10^6 \times 3}{5}$$

$$= 0.6 \times 10^6 \text{ N/m}^2$$

The indicated power of a four stroke engine is given by,

$$IP = \frac{p_m \; L \; A \; N}{60 \times 2 \times 1000}$$

$$= \frac{0.6 \times 10^6 \times 0.2 \times \bullet (0.15)^2 \times 500}{4 \times 60 \times 2 \times 1000}$$

$$= \mathbf{8.83 \text{ kW}}$$

Problem 3.9

Find the indicated power of a four–stroke petrol engine of swept volume of 6 litres and running at 1000 rpm. The mean effective pressure is 600 kN/m².

Solution :

$p_m = 600 \text{ kN/m}^2 = 6 \text{ bar}$ Swept volume $LA = 6 \text{ litres} = 6 \times 10^{-3} \text{ m}^3$

The indicated power of a four stroke engine is given by,

$$IP = \frac{100 \, p_m \; L \; A \; N}{60 \times 2 \times 1000} \text{ kW}$$

$$= \frac{100 \times 6 \times 6 \times 10^{-3} \times 800}{60 \times 2}$$

$$= \mathbf{24 \text{ kW}}$$

Problem 3.10

A four cylinder four–stroke engine running at 1000 rpm develops an indicated power of 15 kW. The mean effective pressure is 5×10^5 N/m². Find the diameter of the cylinder and the stroke of the piston when the ratio of diameter to stroke is 0.8.

Solution :

$IP = 15 \text{ kW}$ $p_m = 5 \times 10^5 \text{ N/m}^2 = 5 \text{ bar}$ $N = 1000 \text{ rpm}$ $\frac{D}{L} = 0.8$

$$\text{Indicated Power developed / cylinder} = \frac{\textit{Total Engine Power}}{\textit{Number of Cylinders}}$$

$$= \frac{15}{4} \text{ kW}$$

$$= \mathbf{3.75\ kW}$$

$$IP = \frac{100\, p_m\ L\ A\ N}{60 \times 2}$$

$$3.75 = \frac{100 \times 5 \times 1.25D \times \bullet D^2 \times 1000}{4 \times 60 \times 2}$$

$$D^3 = 9.167 \times 10^{-4}\ \text{m}^3$$

$$D = 0.09714\ \text{m}$$

$$= \mathbf{97.14\ mm}$$

$$\frac{D}{L} = 0.8$$

$$L = \frac{97.14}{0.8}$$

$$= \mathbf{121.42\ mm}$$

Problem 3.11

A four-stroke petrol engine of 100 mm bore and 150 mm stroke consumes 1 kg of fuel per hour. The mean effective pressure is 7 bar and its indicated thermal efficiency is 30%. The calorific value of the fuel is 40 × 10³ kJ/kg. Find the crankshaft speed.

Solution :

$p_m = 7$ bar $\qquad \bullet_{\text{I Thermal}} = 30\% \qquad CV = 40 \times 10^3$ kJ/kg

$L = 150$ mm $= 0.15$ m $\qquad D = 100$ mm $= 0.1$ m $\qquad m = 1$ kg/hour $= \frac{1}{3600}$ kg/s

For a four–stroke engine, the indicated power is given by,

$$IP = \frac{100\, p_m\ L\ A\ N}{60 \times 2}\ kW$$

$$= \frac{100 \times 7 \times 0.15 \times \bullet (0.1)^2 \times N}{4 \times 60 \times 2}$$

$$= \mathbf{6.87 \times 10^{-3}}\ N\ kW$$

The indicated thermal efficiency is given by,

$$\eta_{B\ Thermal} = \frac{IP}{CV \times m}$$

$$0.3 = \frac{6.87 \times 10^{-3} N}{40 \times 10^{3} \times \frac{1}{3600}}$$

$$N = \mathbf{485.2\ RPM}$$

Problem 3.12

The following data refers to a single cylinder 4 stroke petrol engine.

Cylinder diameter = *20 cm*

Stroke of the piston = *40 cm*

Engine speed = *400 rpm*

Indicated mean effective pressure = *7 bar*

Fuel consumption = *10 litres / hour*

Calorific value of the fuel = *45000 kJ/kg*

Specific gravity of the fuel = *0.8*

Find the indicated thermal efficiency.

Solution :

D = *20 cm* *Fuel Consumption* = *4 litres /hour*

L = *40 cm* *Specific gravity* = *0.8*

N = *400 rpm* *Calorific value of fuel* = *45000 kJ/kg*

p_m = *7 bar* *Calorific value of fuel* = *45000 kJ/kg*

The indicated power of a four–stroke engine is given by,

$$IP = \frac{100\, p_m\ L\ A\ N}{60 \times 2}\ \text{kW}$$

$$= \frac{100 \times 7 \times 0.4 \times \pi (0.2)^2 \times 400}{4 \times 60 \times 2}$$

$$= \mathbf{29.32\ kW}$$

The indicated thermal efficiency is given by,

$$\eta_{B\ Thermal} = \frac{IP}{CV \times m} \times 100$$

$$= \frac{29.32}{45000 \times 10 \times \frac{0.8}{3600}} \times 100$$

$$= \mathbf{29.32\ \%}$$

Problem 3.13

On a single cylinder four–stroke petrol engine, the following readings were taken :

Load on the brake drum = *40 kg*

Spring balance reading = *5 kg*

Diameter of the brake drum = *120 cm*

Fuel consumption = *3 kg/hour*

Calorific value of the fuel = *42000 kJ/kg*

Engine Speed = *500 rpm*

Find the brake thermal efficiency.

Solution :

Net Load on brake durm = (40 – 5)

= 35 kg

Radius of the brake durm = $\frac{120}{2}$ = 60 cm = 0.6m

Torque on the brake durm = $\frac{9.81\ WR}{1000}$ kNm

$$= \frac{9.81 \times 35 \times 0.6}{1000}$$

$$= \mathbf{0.206\ kNm}$$

The Brake Power is given by,

$$BP = \frac{2\pi N T}{60}\ kW$$

$$= \frac{2\pi \times 500 \times 0.206}{60}$$

$$= \mathbf{10.78\ kW}$$

The Brake thermal efficiency is given by,

$$\eta_{B\ Thermal} = \frac{BP}{CV \times m} \times 100$$

$$= \frac{10.78 \times 100 \times 3600}{42000 \times 3}$$

$$= \mathbf{30.8\%}$$

Problem 3.14

A two-stroke diesel engine has a piston diameter of 200 mm and a stroke of 300 mm. It has mean effiective pressure of 2.8 bar and a speed of 400 rpm. The diameter of the brake drum is 1 metre and the effective brake load is 64 kg. Find the indicated power, the brake power, the mechanical efficiency of the engine and the average piston speed.

Solution :

$$\text{Indicated Power} = \frac{100\, p_m\ L\ A\ N}{60}\ \text{kW}$$

$$= \frac{100 \times 2.8 \times \frac{300}{1000} \times \bullet \frac{(0.2)^2}{4} \times 400}{60}$$

$$= \mathbf{17.6\ kW}$$

$$\text{Brake Power} = \frac{2 \bullet \cdot \bullet N\ T}{60}\ \text{kW}$$

$$= \frac{2 \times \bullet \times 400 \times 9.81 \times 64 \times 0.5}{60 \times 1000}$$

$$= \mathbf{13.15\ kW}$$

$$\text{Mechanical Efficiency} = \frac{\textit{Brake Power}}{\textit{Indicated Power}} \times 100$$

$$= \frac{13.15}{17.6} \times 100$$

$$= \mathbf{74.7\ \%}$$

$$\text{Average Piston Speed} = 2\,LN$$

$$= 2 \times 0.3 \times 400$$

$$= 240\ \text{m/min}$$

Problem 3.15

A gas engine working on a four–stroke cycle has a cylinder of 250 mm diameter, length of stroke 450 mm, and is running at 180 rpm. Its mechanical efficiency is 80% when the mean effective pressure is 0.65 MPa. Find (i) indicated power, (ii) brake power and (iii) friction power.

Solution :

$$\text{Indicated Power} = \frac{p_m \; L \; A \; n}{60} \text{ W}$$

$$= \frac{0.65 \times 10^6 \times 0.45 \times \pi \frac{(0.25)^2}{4} \times \frac{180}{2}}{60 \times 1000} \text{ kW}$$

$$= \mathbf{21.53 \text{ kW}}$$

$$\text{Mechanical Efficiency} = \frac{\textit{Brake Power}}{\textit{Indicated Power}} \times 100$$

$$\text{Brake Power} = \frac{\textit{Mechanical Efficiency} \times \textit{Indicated Power}}{100}$$

$$= \frac{80 \times 21.53}{100}$$

$$= \mathbf{17.23 \text{ kW}}$$

$$\textit{Friction Power} = \textit{Indicated Power} - \textit{Brake Power}$$

$$= 21.53 - 17.23$$

$$= \mathbf{4.3 \text{ kW}}$$

Problem 3.16

A single cylinder four–stroke engine has a swept volume of 4.5 litre. The m.e.p. is 0.65 MPa and the engine speed is 505 rpm. If there are 250 explosions per minute and the brake torque is 176 Nm, find indicated power and brake power of the engine.

Solution :

In the indicated power equation, L will be in metre and A will be in sq.m. Therefore the product LA will be in m^3. The product LA is equal to the stroke volume. The stroke volume of 4.5 litre is now expressed in m^3.

$$LA = 4.5 \text{ litre}$$

$$= 4.5 \times 1000 \text{ cm}^3$$

$$= 0.0045 \text{ m}^3$$

Now $$\text{Indicated Power} = \frac{p_m \; L \; A \; n}{60} \text{ W}$$

Where, n = number of cycles per min

$$= \frac{N}{2} \text{ for 4 stroke engine}$$

$$= \frac{505}{2} = 252.5 \text{ cycles per min}$$

But, there are only 250 explosions per min. Although 252.5 cycles are performed every minute, only in 250 cycles firing will take place, the remaining, 252.5 – 250 = 2.5 cycles per min, will be idle cycles. Therefore in the indicated power equation, substitute n = 250 cycles per min instead of 252.5 cycles per min.

$$\text{Indicated Power} = \frac{p_m\ L\ A\ n}{60}\ \text{W}$$

$$= \frac{0.65 \times 10^6 \times 0.0045 \times 250}{60 \times 1000}\ \text{kW}$$

$$= \mathbf{12.18\ kW}$$

$$\text{Brake Power} = \frac{2 \bullet .\ N\ T}{60}\ \text{W}$$

$$= \frac{2 \times \bullet \times 505 \times 176}{60 \times 1000}\ \text{kW}$$

$$= \mathbf{9.3\ kW}$$

Note : In the Brake Power equation N = 505.

Problem 3.17

A single cylinder four–stroke engine runs at 1000 r.p.m. and has a bore of 115 mm and has a stroke of 140 mm. The brake load is 6 kg at 600 mm radius and the machanical efficiency is 80 percent. Calculate brake power and mean effective pressure.

Solution :

$$\text{Brake Power} = \frac{2 \bullet\ N\ T}{60}\ \text{W}$$

$$= \frac{2 \times \bullet \times 1000 \times 6 \times 9.81 \times 0.6}{60}$$

$$= \mathbf{3698.3\ W}$$

$$\text{Mechanical Efficiency} = \frac{\textit{Brake Power}}{\textit{Indicated Power}} \times 100$$

$$\text{Indicated Power} = \frac{\textit{Brake Power}}{\textit{Mechanical Efficiency}} \times 100$$

$$= \frac{3698.3}{80} \times 100$$

$$= \mathbf{4622.9\ W}$$

$$\text{But Indicated Power} = \frac{p_m\ L\ A\ n}{60 \times 2}\ \text{W}$$

$$\therefore \quad p_m = \frac{\textit{Indicated Power} \times 60 \times 2}{L\ A\ N}\ \text{N/m}^2$$

$$= \frac{4622.9 \times 60 \times 2}{\frac{140}{1000} \times \bullet\bullet \frac{(0.115)^2}{4} \times 1000}$$

$$= 381500\ \text{N/m}^2 = \mathbf{3.815\ bar}$$

Problem 3.18

The following observations were obtained during a trial on a four–stroke diesel engine.

Cylinder diameter = *25 cm*
Stroke of the Piston = *40 cm*
Crankshaft speed = *250 rpm*
Brake load = *70 kg*
Brake drum diameter = *2 m*
Mean effective pressure = *6 bar*
Diesel oil consumption = *0.1 m^3/min*
Specific gravity of diesel = *0.78*
Calorific Value of diesel = *43900 kJ/kg*

Find :

1. *Brake Power*
2. *Indicated Power*
3. *Frictional Power*
4. *Mechanical Efficiency*
5. *Brake Thermal Efficiency*
6. *Indicated Thermal Efficiency*

Solution :

1. *Brake Power :*

$$\text{Torque applied} \quad T = \frac{9.81\ WR}{1000} \ \textbf{kNm}$$

$$= \frac{9.81 \times 70 \times 1}{1000} \ \textbf{kNm}$$

$$= 0.686 \ \text{kNm}$$

$$\text{Brake Power} = \frac{2 \bullet N T}{60}$$

$$= \frac{2 \bullet \times 250 \times 0.686}{60 \times 1000}$$

$$= \textbf{17.95 kW}$$

2. *Indicated Power :*

$$\text{Indicated Power} = \frac{100\ p_m\ L\ A\ N}{60 \times 2} \ \textbf{kNm}$$

$$= \frac{100 \times 6 \times 0.4 \times \bullet\ (0.25)^2 \times 250}{4 \times 60 \times 2}$$

$$= \textbf{24.54 kW}$$

3. *Frictional Power :*

$$\text{Frictional Power} = \text{Indicated Power} - \text{Brake Power}$$
$$= 24.54 - 17.95$$
$$= \mathbf{6.59\ kW}$$

4. *Mechanical Efficiency :*

$$\eta_{Mech} = \frac{\text{Brake Power}}{\text{Indicated Power}} \times 100$$
$$= \frac{17.95}{24.54} \times 100$$
$$= \mathbf{73.14\%}$$

5. *Brake Thermal Efficiency :*

$$\eta_{B.Thermal} = \frac{\text{Brake Power}}{CV \times m} \times 100$$
$$= \frac{17.95}{43900 \times \frac{0.1 \times 0.78}{60}} \times 100$$
$$= \mathbf{31.45\%}$$

6. *Indicated Thermal Efficiency :*

$$\eta_{I.Thermal} = \frac{24.54}{43900 \times \frac{0.1 \times 0.78}{60}} \times 100$$
$$= \mathbf{43\%}$$

REFRIGERATION AND AIR CONDITIONING

3.16 Introduction

From earlier times, the art of *artificial cooling* is employed in ice making, preservation of the perishables such as, milk, food, drinks, medicines, etc. and indoor air cooling to provide human comfort and cool environment required in electronic, precision manufacturing and process industries, and also in numerous other applications like cryogenics, etc. The two methods employed for artificial cooling are ; (i) *refrigeration* and (ii) *air-conditioning*. Refrigerators are used in ice making, preservation of perishables, cryogenics, etc. and air-conditioners are used in indoor air cooling.

3.17 Refrigeration Defined

Refrigeration is defined as a method of reducing the temperature of a system below that of the surroundings and maintains it at the lower temperature by continuously abstracting the heat from it.

3.18 Principle of Refrigeration

In refrigeration, the heat is to be removed continuously from a system at a *lower temperature* and transfer it to the surroundings at a *higher temperature*. This operation according to the Second Law of Thermodynamics can only be performed by the aid of the external work. Therefore in a